

Near-Symbol-Rate Linear Least-Square Longitudinal Power Monitoring

Junho Chang, Choloong Hahn, Qingyi Guo, and Zhiping Jiang*

Ottawa Research Centre, Huawei Technologies Canada, 303 Terry Fox Drive, Ottawa, Canada,
zhiping.jiang@huawei.com

Abstract We characterize the dependency of the spatial response function on the waveform sampling rate used for LPM. By proposing an optimized near-symbol-rate approach, we achieve mean absolute error of the estimated profile comparable to 2-SpS processing with lower sampling requirements, supporting more resource-efficient real-time monitoring. ©2026 The Author(s)

Introduction

In modern optical networks, accurate estimation of distributed power evolution provides enhanced visibility into nonlinear distortion (NLD), span conditions, and system impairments, thereby enabling intelligent system control and optimization. In this context, longitudinal power monitoring (LPM) has emerged as an essential technique for optical system provisioning and network management [1-5]. Between the two primary approaches, linear least-square (LLS)-based LPM is particularly attractive because it resolves convolutional spatial ambiguity and enables calibration-free absolute power estimation [6-8]. However, LLS requires inversion of a convolution matrix whose dimensions scale with both the number of spatial steps and waveform samples, imposing significant computational and memory demands [9]. While increasing the spatial step size can reduce implementation complexity, excessively large steps introduce substantial inaccuracies [10]. Furthermore, the optimal step size decreases as the signal baud rate increases, making conventional LLS-based LPM increasingly impractical for high-baud-rate and/or long-haul systems. Although analytical formulations have been proposed as alternatives [11], they typically rely on restrictive assumptions. Another promising strategy for reducing complexity is symbol-level LPM utilizing waveforms at one sample per symbol (SpS). While 2 SpS is common in prior LPM works to accurately capture NLD transients, symbol-rate or near-symbol-rate data (e.g., post-equalization symbols or 1.25-SpS fractionally sampled data) are often more readily available within existing DSP architectures. Forcing a 2-SpS rate for LPM thus incurs additional overhead in memory and computation. Although symbol-rate LPM has been explored [12-13], it reportedly suffers from noticeable inaccuracies compared to higher-SpS data, even after calibration [12].

In this paper, we investigate the root cause of this inaccuracy in LLS-based LPM under near-symbol-rate sampling, including 1-SpS operation.

We show that the spatial response function (SRF) effectively broadens at lower sampling rates, whereas conventional LLS formulations implicitly assume a narrower SRF corresponding to higher sampling rates (e.g., 2-SpS). This SRF mismatch leads to under-deconvolution in the LLS process and, consequently, biased power profile slopes. Based on this insight, we propose a modified LLS algorithm optimized for near-symbol sampling. By accurately modelling and incorporating the true SRF into the convolution matrix, the proposed method significantly reduces estimation errors. Furthermore, we show that the broadened SRF increases the minimum viable spatial step size in LLS-based LPM, allowing for a further reduced matrix size while maintaining accuracy.

Impact of Down-sampled Waveforms on LPM

In a dispersion unmanaged optical link, the received waveform at Rx can be modelled as [14]: $A(L, t) = u(L, t) + \int_0^L -j\gamma P(z) \Delta u_z(L, t) dz$, where $u(L, t)$ is the linearly propagated pure-signal waveform at Rx (i.e., $z = L$), γ is the fibre nonlinear coefficient, $P(z)$ is the signal power at location z , and $\Delta u_z(L, t)$ is the NLD waveform generated at location z . For LPM, a reference NLD waveform is typically reconstructed at each location z_k as $\Delta \tilde{u}_{z_k}[L, n] \equiv -j\hat{D}_{z_k, L} \hat{N}_{\text{eRP}} \hat{D}_{0, z_k} u(0, nT/N_{\text{LPM}})$, using a nonlinear operator such as enhanced regular perturbation (eRP) model. Here, $u(0, nT/N_{\text{LPM}})$ is the discrete Tx waveform sampled at N_{LPM} SpS, where T denotes the symbol duration. When $N_{\text{LPM}} = 1$, i.e., 1 SpS, the post-decision symbols can be directly used. The operators $\hat{D}_{z_1, z_2}$ and $\hat{N}_{\text{eRP}} \equiv (|\cdot|^2 - 2(|\cdot|^2))(\cdot)$ represent the chromatic dispersion (CD) from z_1 to z_2 and the nonlinear transformation, respectively. In contrast, the actual NLD waveform generated during propagation is given by $\Delta u_{z_k}(L, t) = -j\hat{D}_{z_k, L} \hat{N}_{\text{eRP}} \hat{D}_{0, z_k} u(0, t)$. After down-sampling within the Rx DSP, this becomes $\Delta u_{z_k}[L, n] = \Delta u_{z_k}(0, nT/N_{\text{LPM}})$. Following the methodology in [14], we construct the actual and reference NLD matrices as:

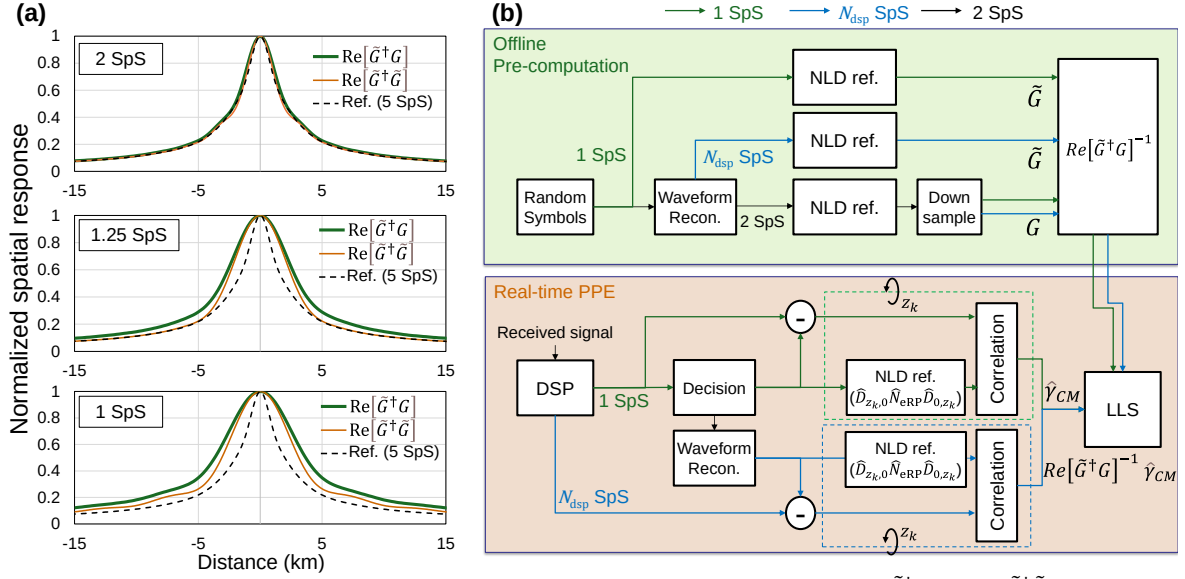


Fig. 1: (a) Normalized SRFs for different sampling rates obtained using correlation terms, $\text{Re}[\tilde{G}^\dagger G]$ and $\text{Re}[\tilde{G}^\dagger \tilde{G}]$, (solid) compared to sampling-independent SRFs (dashed). (b) Schematic diagrams of proposed LPM method.

$$G = [\Delta u_{z_0}[L, n], \dots, \Delta u_{z_k}[L, n], \dots, \Delta u_{z_M}[L, n]]$$

$$\tilde{G} = [\Delta \tilde{u}_{z_0}[L, n], \dots, \Delta \tilde{u}_{z_k}[L, n], \dots, \Delta \tilde{u}_{z_M}[L, n]]$$

where M is the total number of LPM steps. The received NLD component is represented by $G\gamma'$, with $\gamma' = \gamma^P(z)$. The output of correlation-method (CM) LPM [8] is $\hat{\gamma}'_{\text{CM}} = \text{Re}[\tilde{G}^\dagger G\gamma']$, while the LLS solves for γ' as: $\hat{\gamma}'_{\text{LLS}} = (\text{Re}[\tilde{G}^\dagger \tilde{G}])^{-1} \text{Re}[\tilde{G}^\dagger G\gamma']$. The inherent challenge in this estimation is the mismatch between the reconstructed and actual NLD matrices, such that $\tilde{G}^\dagger \tilde{G} \neq \tilde{G}^\dagger G$.

We first evaluate $\text{Re}[\tilde{G}^\dagger \tilde{G}]$ and $\text{Re}[\tilde{G}^\dagger G]$ for different LPM sampling rate, N_{LPM} . Each row corresponds to the SRF at a given location. In this evaluation, a 75-Gbaud dual-polarization (DP) 16QAM signal was simulated over a standard single-mode fibre (SSMF) using the split-step Fourier method (SSFM). The normalized SRFs are illustrated in Fig. 1(a) for various N_{LPM} . The SRF obtained with negligible sampling effect (i.e., $N_{\text{LPM}} = 5$) is also shown as a reference in dashed line. When $N_{\text{LPM}} = 2$, the two matrices are nearly identical and reference as well, showing there is little impact of sampling rate. However, as the sampling rate approaches 1 SPS, the SRF not only broadens but also deviates from the ideal response in both matrices. Notably, broadening of $\text{Re}[\tilde{G}^\dagger G]$ is more pronounced than that of $\text{Re}[\tilde{G}^\dagger \tilde{G}]$, leading to insufficient deconvolution when using the conventional formulation. This explains the large estimation errors—especially near amplifier locations—reported in prior 1-SPS studies [12].

To address this issue, we propose using $\text{Re}[\tilde{G}^\dagger G]$ directly within the LLS formulation. The schematic diagram for this approach is illustrated

in Fig. 1(b). In this architecture, the real-time block consists solely of the implementation-friendly CM-based LPM. The more computationally intensive LLS matrix calculation is shifted to an offline process, a strategy similar to that employed in [13]. For the correlation step, the system utilizes data only at or near the symbol rate, which is readily accessible from standard DSP. This bypasses the need for the upsampling processes, additional computational overhead, and high memory typically required for NLD reference generation. Depending on hardware accessibility, the input can be: 1) symbols immediately prior to decision (i.e., 1 SPS), or 2) DSP-rate waveforms sampled at the internal DSP rate (i.e., N_{DSP} SPS). Note that the CM is inherently parallelizable—unlike the matrix inversion required for LLS—ensuring it does not impose heavy memory constraints on the real-time hardware. The LLS reconstruction matrix is pre-computed offline, as shown in the upper block of Fig. 1(b). We can utilize emulated symbols or waveforms to construct both G and $\tilde{G}$ matrices, subsequently calculating $(\text{Re}[\tilde{G}^\dagger \tilde{G}])^{-1}$. This pre-calculated operator is then applied to the real-time CM results.

Performance Evaluation

We numerically evaluated the performance of the proposed method using a 138-Gbaud DP-16QAM signal transmitted over 80-km SSMF spans. The span launch power was set to 8 dBm, and optical amplifier noise was neglected to isolate the deconvolution capabilities of the LLS formulation. The fibre parameters—dispersion, attenuation, and nonlinear coefficient—were 17 ps/nm/km, 0.2 dB/km, and $1.4 \text{ W}^{-1}\text{km}^{-1}$, respectively. At the receiver, digital CD compensation and carrier phase recovery were performed on

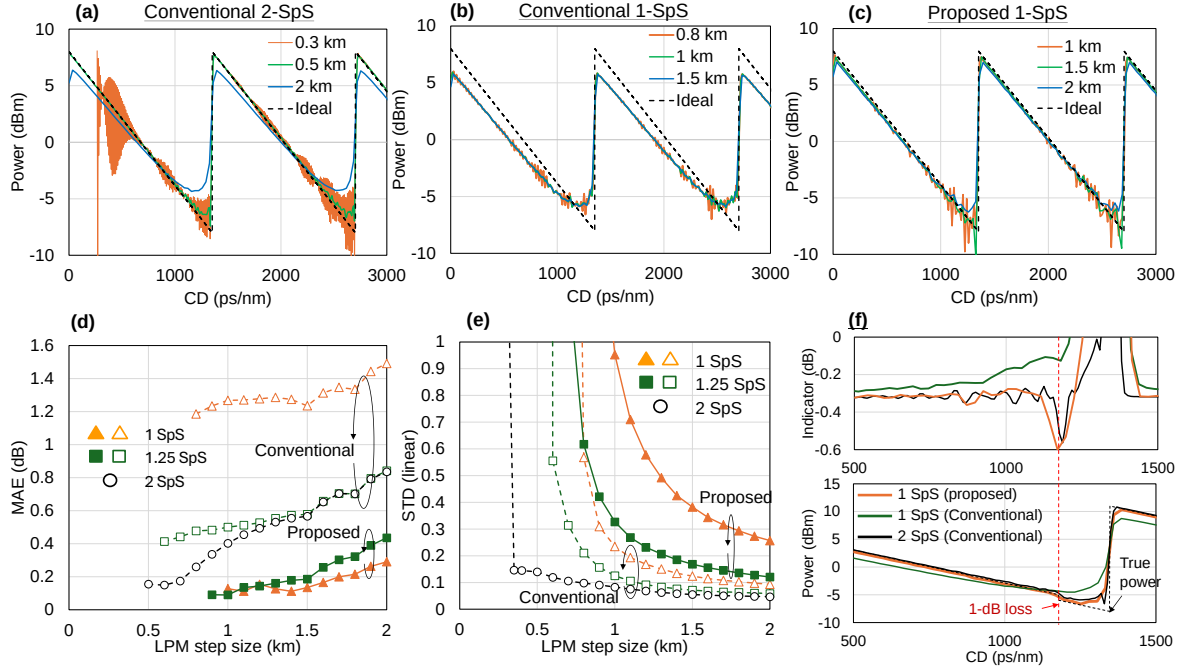


Fig. 2: (a) Power profiles estimated by conventional 2-SpS, (b) conventional 1-SpS, and (c) proposed 1-SpS LLS-LPM for varying step sizes. (d) MAE and (e) STD of the estimates as a function of step size for different N_{LPM} (32 ksymbol/set) (f) Estimated profiles featuring a 1-dB loss anomaly and the corresponding loss indicator derived from spatial derivatives.

waveforms sampled at 2 SpS. These waveforms were subsequently resampled to N_{LPM} values of 2, 1.25, or 1 SpS. The LPM was then executed according to the methodology described in the previous section. Fig. 2(a-c) illustrates the estimation results. In the conventional 2-SpS case, the true power profile was accurately estimated only when the step size was optimized (~ 0.5 km in this condition). An excessively large step size resulted in incomplete deconvolution, while an overly small step size introduced numerical instability [6]. When employing conventional 1-SpS approach (utilizing $\text{Re}[\tilde{G}^\dagger \tilde{G}]$), the estimation consistently exhibited inaccuracies of up to 3 dB, particularly at amplifier locations, regardless of the chosen step size. In contrast, by using the proposed 1-SpS LPM with $\text{Re}[\tilde{G}^\dagger G]$, this inaccuracy was virtually eliminated, showing excellent alignment with the actual power profile across the entire link. A similar improvement was observed for the 1.25-SpS case, though the initial deviation was less severe.

Interestingly, the optimal step size for 1-SpS case is larger than that for the 2-SpS, with the proposed method requiring a slightly coarser step size than the conventional 1-SpS approach. To quantify these impacts, we calculated the mean absolute error (MAE) and standard deviation (STD) of the estimated profiles across varying parameters, as shown in Fig. 2(d-e). While the conventional 1-SpS method exhibited significant MAE across all tested step sizes, the conventional 2-SpS case achieved an MAE below 0.2 dB only when the step size was maintained under 0.7 km. In contrast, the proposed

1-SpS and 1.25-SpS methods maintained low MAE levels even with a much coarser step size of ~ 1.5 km, while keeping the STD reasonably low. Nevertheless, the proposed method still exhibited higher fluctuations; the STD for the 1-SpS and 1.25-SpS cases was approximately 1.5 to 2.5 times higher than that of the 2-SpS case at their respective optimal step sizes. Although this increased variance implies a greater requirement for averaging to match the precision of 2-SpS, the advantages remain significant—specifically, the improved estimation accuracy near abrupt power transitions. Fig. 2(f) illustrates an example of a 1-dB anomaly loss detected 10 km before the amplifier. These results clearly demonstrate that the proposed method offers significantly better detectability than conventional 1-SpS approaches, providing a robust balance between computational efficiency and monitoring performance.

Conclusions

We have established the link between waveform sampling rates and the SRFs of LLS-based LPM. Our findings reveal that the inaccuracies previously associated with 1-SpS operation are not fundamental limitations, but stem from a SRF mismatch. By integrating a corrected modelling approach, we achieved near-symbol-rate performance comparable to that of oversampled benchmarks. This shift to a coarser spatial and temporal grid drastically lowers the barriers to hardware implementation, offering a monitoring solution that fits within the strict memory and power constraints of modern transceivers.

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